Project Submission

Group 6 | Group Leader: Shashank Sadashiv

Team Members

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Assignment Part 1: Base Research Paper

SED: A Simple Encoder-Decoder for Open-Vocabulary Semantic Segmentation
Paper
https://github.com/xb534/SED

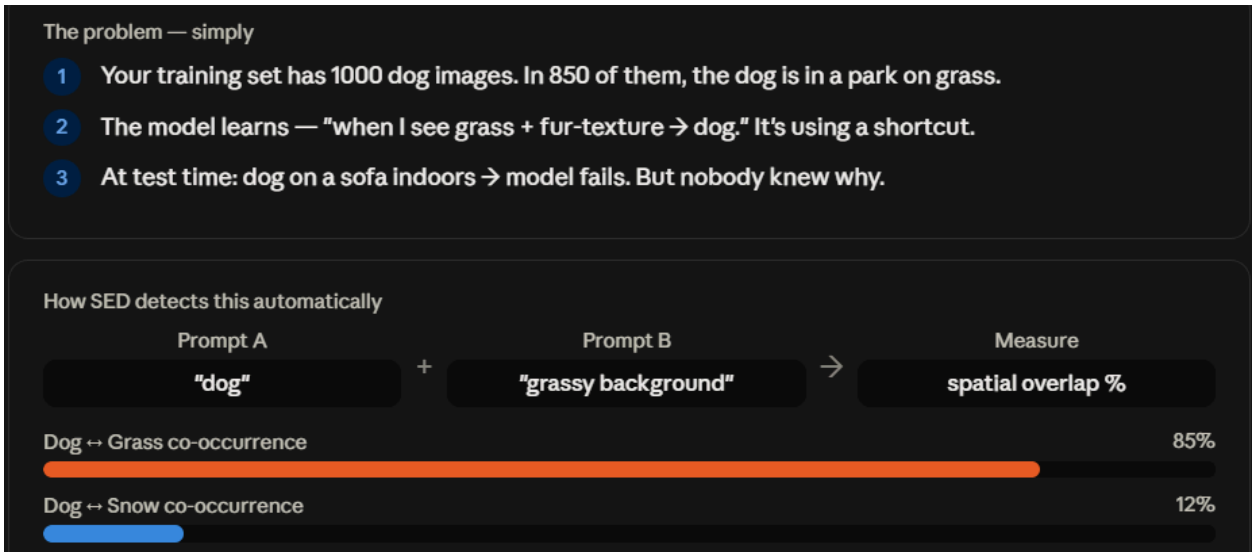
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024

Abstract

Open-vocabulary semantic segmentation strives to distinguish pixels into different semantic groups from an open set of categories. Most existing methods explore utilizing pre-trained vision-language models in which the key is to adopt the image-level model for pixel-level segmentation task. In this paper we propose a simple encoder-decoder named SED for open-vocabulary semantic segmentation which comprises a hierarchical encoder-based cost map generation and a gradual fusion decoder with category early rejection. The hierarchical encoder-based cost map generation employs hierarchical backbone instead of plain transformer to predict pixel-level image-text cost map. Compared to plain transformer hierarchical backbone better captures local spatial information and has linear computational complexity with respect to input size. Our gradual fusion decoder employs a top-down structure to combine cost map and the feature maps of different backbone levels for segmentation. To accelerate inference speed we introduce a category early rejection scheme in the decoder that rejects many no-existing categories at the early layer of decoder resulting in at most 4.7 times acceleration without accuracy degradation. Experiments are performed on multiple open-vocabulary semantic segmentation datasets which demonstrates the efficacy of our SED method.

Assignment Part 2: Proposed Research Task

A key pain point in ML research is detecting spurious visual correlations in training datasets, for example, a 'dog detector' that secretly learns to detect grass because dogs appear on grass 85% of the time. We automate this bias-detection task using SED: by independently segmenting pairs of semantic concepts across the entire training set and computing their spatial co-occurrence statistics, we can surface concept entanglements before training begins. This transforms a months-long post-hoc debugging task into an automated pre-experiment report, using SED's open-vocabulary capability as an analysis instrument rather than a segmentation output.



SED can segment "X" and "Y" independently. If segmenting concept X consistently also captures concept Y, you've discovered a spurious visual correlation in your dataset that would confound experiments.

- E.g., your "person" segments always include "grass" → your model may be learning background bias
- No retraining needed — just sweep text prompts over your dataset
- **Researcher task automated**: *Pre-experiment dataset sanity checking*